

INTERNSHIP REPORT

On

RF Sensitivity Analysis with Cold- Test Measurement for Folded Waveguide TWT Structures with Material Choices

Submitted By:

Piyush Gupta

Roll Number: EE22B056

Bachelor of Technology (B.Tech)

Department of Electrical Engineering

Indian Institute of Technology, Tirupati

Internship Duration:

12 June 2025 – 24 July 2025



Project Guides:

7 ,



Microwave Tube Research and Development Centre (MTRDC)

Defence Research and Development Organisation (DRDO)

Jalahalli, Bangalore



सूक्ष्मतरंग नलिका अनुसंधान तथा विकास केन्द्र

रक्षा अनुसंधान तथा विकास संगठन (डीआरडीओ), रक्षा मंत्रालय, भारत सरकार

भारत इलेक्ट्रॉनिक्स उत्तर द्वार के पास, जालहल्ली, बेंगलूरु- 560013

Microwave Tube Research and Development Centre

DRDO, Ministry of Defence, Government of India

Near Bharat Electronics North Gate, Jalahalli, Bangalore-560013, Karnataka

CERTIFICATE

This is to certify that the internship report entitled 'RF Sensitivity Analysis with Cold-Test Measurement for Folded Waveguide TWT Structures with Material Choices' is a record of work carried out by Shri Piyush Gupta at Microwave Tube Research and Development Centre (MTRDC), Defense Research and Development Organization (DRDO), Ministry of Defense, in partial fulfilment of requirement for the award of **Bachelor Of Engineering in Electrical Engineering** of 'Indian Institute of Technology Tirupati', under my guidance and supervision. During this period 12th June 2025 to 24th July 2025, he has carried out good work and completed the same in stipulated time.

U. Santra
24-07-25

Dr MRITYUNJAY SANTRA,
OS & Scientist 'H'
Head, HRD Training Division

A. Srivastava
24/07/2025

Signature of Guide
Dr. ANURAG SRIVASTAVA,
Scientist 'F'

ABSTRACT

The growing demand for high-frequency, high-power RF sources for applications in defense communication, radar, and imaging systems has renewed interest in vacuum electronic devices (VEDs), particularly Traveling-Wave Tubes (TWTs). This report presents the design, fabrication, simulation, and experimental validation of a Folded Waveguide Traveling-Wave Tube (FWTWT) structure operating in the W-band (75–110 GHz). A detailed study of slow-wave structures (SWS) is carried out, highlighting the advantages of folded waveguides over conventional helices and coupled cavities at sub-THz frequencies, especially in terms of fabrication feasibility, thermal handling, and power output.

The Folded waveguide slow-wave structure (SWS) is one of the most important parts of the TWT. The structure RF sensitivity analysis in terms of mainly S-parameters study have been performed using MWS CST solver. After that, I was involved in the fabrication process which was investigated using two materials—Brass and Aluminum 6061—via CNC machining, and surface quality was analyzed using Surftest and VMM tools. Burr formation, surface roughness, and structural integrity were compared, laying the groundwork for eventual implementation using OFHC copper. The simulated results were validated experimentally through cold-test S-parameter measurements using a Precision Network Analyzer (PNA) with WR-10 extenders.

The comparison of simulated and measured S-parameters (S_{11} , S_{21}) enabled insight into fabrication-induced losses and the skin effect at high frequencies. The study also incorporates rationale for choosing W-band based on evolving spectrum policy and defense requirements. The findings demonstrate the viability of robust brass/aluminum prototyping for early design stages and set a roadmap toward precision OFHC fabrication with diffusion bonding and optimized surface finishing for final deployment.

Future work includes modeling of machining tolerances, diffusion-bonded interface mismatches, and beam–wave interaction studies for full tube integration.

Acknowledgment

The internship opportunity I had at **Microwave Tube Research and Development Centre (MTRDC), DRDO, Bangalore**, has been a significant step in my academic and professional journey. It provided me with valuable insights into interdisciplinary research bridging electrical and mechanical engineering, and I am deeply grateful for the mentorship and support I received.

I am especially thankful to **Mr. Santosh Kumar Jha (Scientist-G)** for his constant guidance, technical clarity, and for helping me bridge the gap between my electrical background and the mechanical domain. I also sincerely thank **Dr. Anurag Shrivastava (Scientist-F)** for his invaluable input, encouragement, and direction throughout my project.

I would like to express my gratitude to **Dr. Abhishek Jha**, Associate Professor, Department of Electrical Engineering, **IIT Tirupati**, for his mentorship, which enabled me to approach this internship confidently and complete the tasks efficiently. I am also grateful to **Dr. N.**

Venkaiah, Professor, Department of Mechanical Engineering, **IIT Tirupati**, for facilitating and approving my participation in this project.

I acknowledge the support of **Mr. Venkateswarlu Y (Scientist-D)** for creating a friendly and motivating work environment, and **Mr. Kartik H.S. (JRF)** for his helpful assistance during the course of the internship. I also thank **Smt. Arut Payai S.K. (Technical Officer-D)** for her guidance and the opportunity to learn from practical microwave experiments.

Finally, I express my sincere thanks to **Dr. Subrata Kumar Datta (Scientist-H)**, Centre Head, MTRDC, for providing all necessary infrastructure and the opportunity to be part of this prestigious organization.

This internship has been a truly transformative experience, and I look forward to applying the knowledge and skills gained to my future endeavors.

ABOUT THE ORGANIZATION

The Microwave Tube Research & Development Centre (MTRDC), established in 1984 under DRDO, Ministry of Defence, specializes in the research, design, and development of advanced microwave tubes, including TWTs, cathodes, and microwave power modules, for strategic applications in radar, electronic warfare, and communications. The center's work spans multiple disciplines such as microwave engineering, high-vacuum technology, brazing, high-voltage engineering, and materials science, supported by state-of-the-art facilities and strong collaborations with production agencies like Bharat Electronics Ltd., DRDO system labs (LRDE, DLRL, DARE, DEAL, RCI, TBRL), and academic/research institutes including IISc, BHU, CEERI, and BARC, aiming for self-reliance in this critical technology sector.

Table of Contents

1. **Acknowledgements**
2. **Abstract**
3. **About the Institute**
4. **Concept of the Project**
5. **Technology Background**
 - 7.1 Principles of Traveling-Wave Tubes (TWTs)
 - 7.2 Slow-Wave Structures (SWS)
 - 7.3 Folded-Waveguide Circuits
6. **Design and Fabrication of the FWTWT**
7. **Simulation Methodology (CST Studio Suite)**

- 9.1 Eigenmode Analysis Setup
 - 9.2 Time-Domain Analysis Setup
 - 8. **Experimental S-Parameter Analysis of FWSWS**
 - 9. **Results and Discussion**
 - 10. **Conclusions and Future Work**
-

List of Figures

- **Figure 1** – Basic Configuration of a Folded Waveguide SWS
 - **Figure 2** – Sectional View of a Single Period FWSWS
 - **Figure 3** – CST Time-Domain Model of Brass and Aluminium FWTWT Structures
 - **Figure 4** – S-Parameter Measurement Setup with PNA and WR-10 Extender
 - **Figure 5** – Comparison of Simulated and Measured S11/S21 for Brass and Aluminium
 - **Figure 6** – Effect of Surface Roughness on S21 (Insertion Loss)
 - **Figure 7** – Machined Brass and Al6061 Structures Showing Burrs in Aluminium
-

List of Tables

- **Table 1** – Normalized Dimensions of FW Structure
 - **Table 2** – Parametric Sweep Results for Phase Velocity Matching
 - **Table 3** – Summary of S-Parameter Measurements and Comparison with Simulation
-

List of Abbreviations

- **TWT** – Traveling-Wave Tube
- **SWS** – Slow-Wave Structure

- **FWTWT** – Folded Waveguide Traveling-Wave Tube
- **FWG** – Folded Waveguide
- **RF** – Radio Frequency
- **CST** – Computer Simulation Technology
- **PNA** – Precision Network Analyzer
- **S11** – Return Loss Parameter
- **S21** – Insertion Loss Parameter
- **OFHC** – Oxygen-Free High Conductivity Copper
- **WR-10** – Waveguide Rectangular Size for 75–110 GHz Band

Section II

Problem Definition – Scope You Worked On

The project aimed to evaluate the feasibility and performance of Folded Waveguide Traveling-Wave Tube (FWTWT) structures in the W-band (90–97 GHz), intended for use in future defense communication systems. The core scope included both mechanical and RF characterization of FWTWTs using CNC-machined structures fabricated from Brass and Al6061, followed by S-parameter validation and simulation modeling.

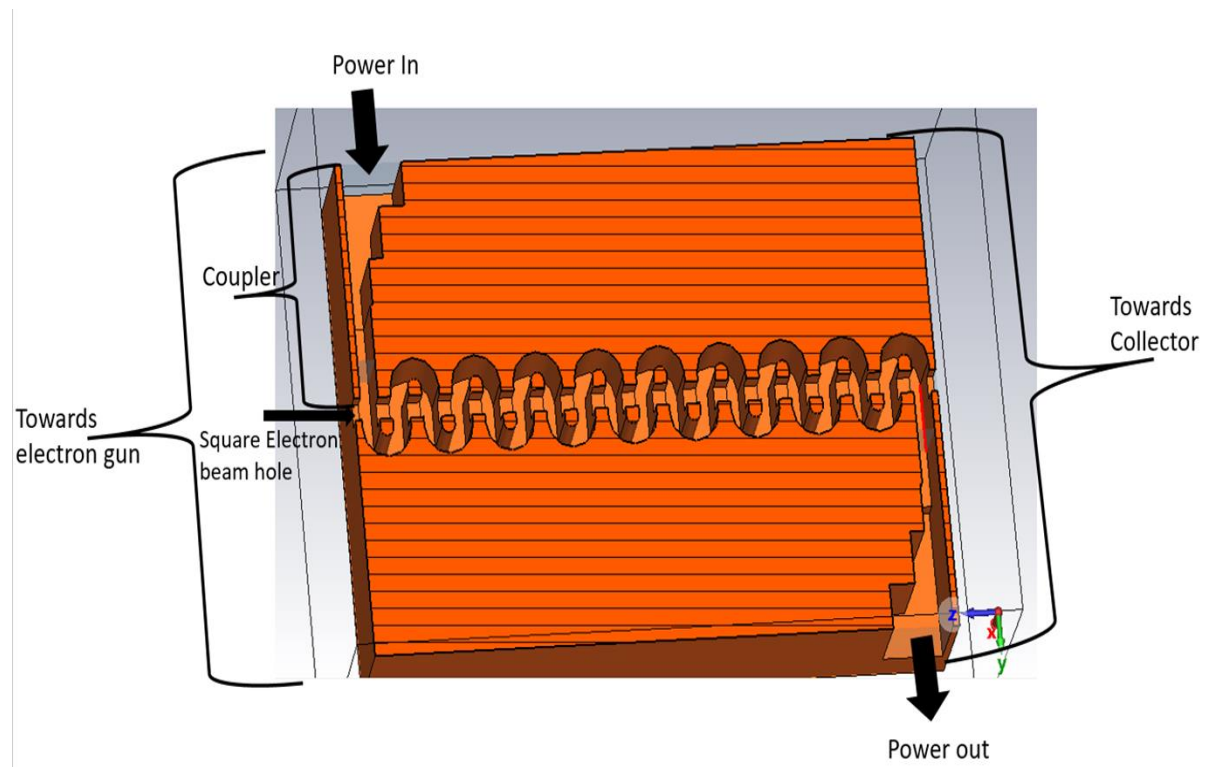
The challenges included verifying whether conventional CNC techniques could meet the surface roughness and tolerance criteria for W-band applications, understanding S-parameter behavior through simulations and measurements, and building the foundation for future OFHC-based fabrication with advanced bonding and finishing processes.

3.3 Folded Waveguide Design for High-Frequency TWTs

Folded waveguide SWS (FWSWS) is chosen for applications such as the 220 GHz BWO due to its robustness, high beam-wave coupling, and compatibility with microfabrication. The FWG structure consists of:

- A periodic folded path for wave propagation
- A square electron beam channel
- An output port for RF extraction

The device operates in the backward-wave mode where group velocity is opposite to the electron beam direction, enabling energy transfer via the -1 space harmonic.



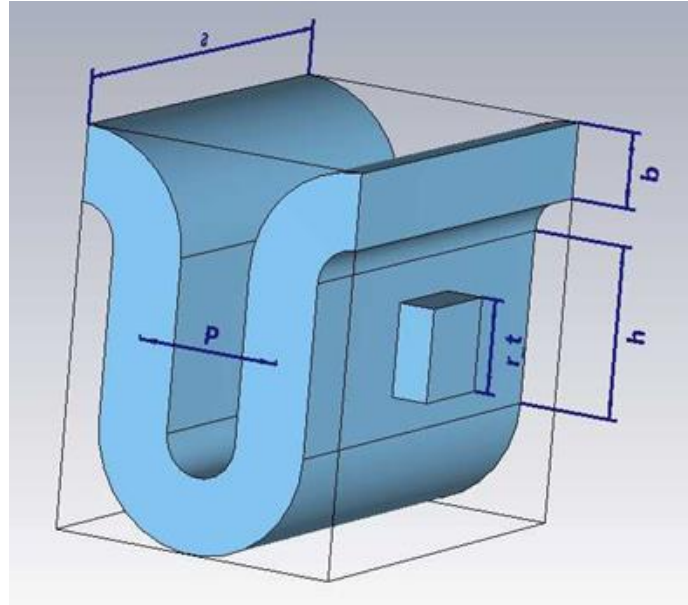
A Folded Waveguide Structure

3.4 Design Parameters

Structural parameters	Symbols
The width of transverse rectangular waveguide	a
The height of transverse rectangular waveguide	b
The pitch length	p
The straight waveguide length	L_0
The diameter of beam tunnel	D
Vertical segment	h
Beam channel side	L_{tube}
Mean radius	R

Table 3 Designed structural parameters.

The beam voltage is selected to synchronize with the phase velocity of the -1 harmonic. For a beam voltage of 10 kV and phase shift of 2.5π , normalized dimensions are chosen to ensure beam-RF synchronization.



Single Period Folded Waveguide SWS

Approach

Current

Traditionally, helix and coupled cavity-based slow-wave structures (SWS) face significant performance limitations in the sub-THz regime due to high RF losses, complex miniaturized fabrication, and thermal constraints. Folded waveguides offer advantages in these domains due to their planar structure, robust design, and ease of alignment. However, validating their mechanical and RF integrity at W-band required a systematic workflow combining fabrication, modeling, and cold-test measurement.

Your Contribution

- Mechanical Domain:
 - Designed the FWTWT structure in two symmetric halves to accommodate beam tunnel limitations.

- Selected and machined two materials: Brass and Al6061. Burr formation and surface quality were compared using VMM and Surftest tools.
- Observed that Aluminum showed noticeable burrs, whereas Brass showed smoother finish. This insight guided further material selection.
- Gained practical understanding of CNC milling constraints for sub-THz components.
- Simulation and RF Domain:
 - Carried out extensive modeling using CST Studio Suite:
 - Eigenmode Solver: Used to tune periodicity for phase velocity matching.
 - Time-Domain Solver: Analyzed S11 and S21 over a roughness sweep to evaluate insertion and return losses.
 - Fabricated structures were cold-tested using a Vector Network Analyzer (Keysight PNA N5227A) with WR-10 extenders, and results were compared against simulation data.
 - Identified the impact of roughness and flange mismatch on the insertion loss performance.
 - Proposed a roadmap for precision OFHC fabrication with diffusion bonding and polished mating surfaces to reduce skin effect losses.

4.3 Material Selection and Machinability Assessment

Two materials were considered:

- **Aluminum Alloy (Al-6061):** Chosen for its light weight, ease of machining, and good conductivity.
- **Brass:** Selected for superior conductivity and surface finish, though heavier and comparatively harder.

Both materials were machined using conventional CNC milling and uncoated carbide tools. No surface polishing or conductive coating (such as silver or gold) was applied after fabrication. During post-fabrication inspection:

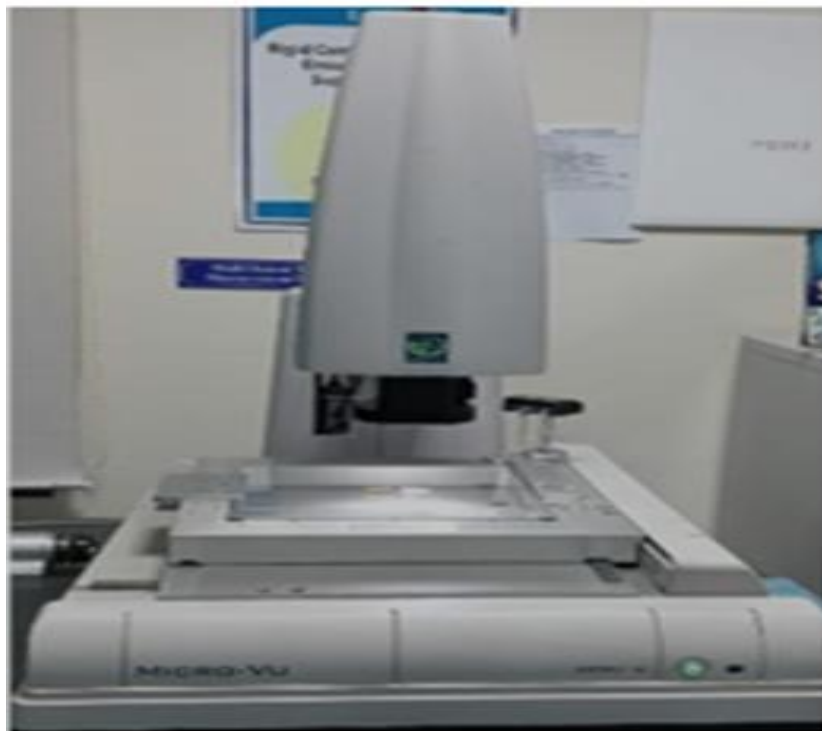
- **Aluminum** exhibited **burr formation**, especially around tunnel and coupling corners.
- **Brass** showed better **machinability and dimensional consistency** but required more care during tool operation due to tool wear.

The choice of uncoated carbide tools allowed micro-features to be milled with minimal tool wear and dimensional deviation. However, for future development, consideration of nano-CNC milling, PCD or titanium-coated tools may help achieve better surface roughness and precision.

4.3.1 Visual Inspection and Surface Finish Measurements

The machined parts were inspected using **VMM (Video Measuring Microscope)** and **SURTEST surface roughness tester** to assess dimensional accuracy and surface quality. Visual comparison of brass and Al6061 structures confirmed the following:

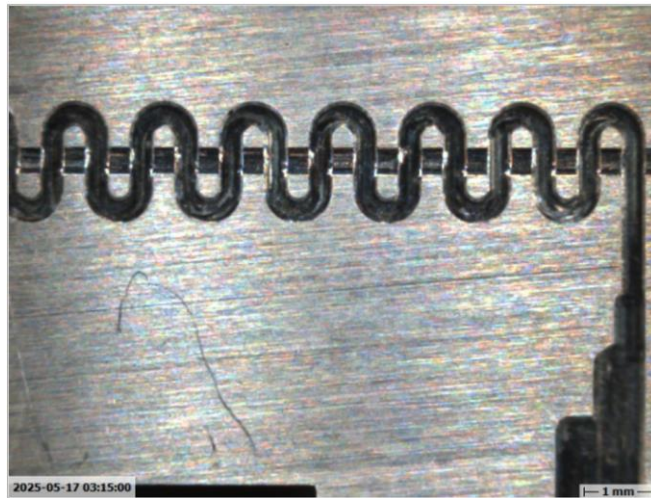
- **Aluminum** parts showed **burr formation** around coupling junctions and waveguide walls.
- **Brass** parts exhibited **cleaner edges** and **no noticeable burrs**, making it more suitable for CNC prototyping of folded waveguide structures.



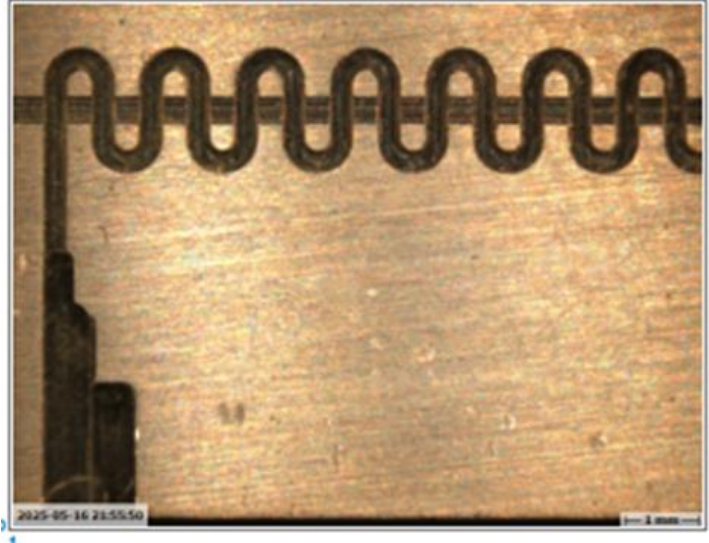
VMM image showing dimensional profiling of waveguide.



SURTEST surface roughness tester setup.



Burr formation in Aluminum sample



Brass structure with clean finish

“This comparative study confirms brass as a more favorable material for prototype development using conventional CNC machining due to reduced burr formation and better edge retention.”

6. EXPERIMENTAL VALIDATION AND S-PARAMETER MEASUREMENT

To validate the simulation results obtained from CST Studio Suite, experimental S-parameter measurements were conducted on the fabricated Folded Waveguide Traveling-Wave Tube (FWTWT) prototypes using a high-precision vector network analyzer setup.

6.1 Experimental Setup and Instrumentation

Cold-test measurements were carried out using the following instrumentation:

- **Vector Network Analyzer (VNA):** Agilent N5227A (10 MHz – 67 GHz)
- **VNA Frequency Extenders:** WR-10 Band modules (frequency range: **75–110 GHz**)
- **Calibration Kit:** Keysight W11644A mechanical kit (for **SOLT calibration**) adapted for WR-10 waveguide ends
- **Waveguide-to-Coax Adapters:** PTC-15VF-01 (Ducommun)
- **Coaxial Cables:** SC-185-MF-24 (Maury Microwave)

Prior to measurement, a **calibration routine** was performed using the **Short-Open-Load-Through (SOLT)** method, ensuring accurate reference planes at the waveguide flanges. This minimized mismatch errors and phase offset issues.

Note: Calibration was a critical step due to the high sensitivity of W-band frequencies to connector discontinuities and flange misalignments.

6.2 Description of Measured Structures

- Structures were fabricated in **two materials**:
 - **Brass** (known for its superior surface finish and conductivity)
 - **Al6061** (lightweight, but observed to develop burrs post-machining)
- Each structure comprised full-length folded waveguides, milled in **two halves**, and assembled using **precision dowel pins and screws**.
- No post-polishing or surface coatings were applied.

6.3 Understanding S-Parameters for FWTWT Cold Test

- **S21 – Insertion Loss**
Indicates forward transmission. Lower S21 (i.e., more negative dB) represents higher total attenuation in the waveguide due to material losses, surface roughness, and radiation.
- **S11 – Return Loss**
Indicates reflected power at input due to impedance mismatch. A deep dip (more negative) implies better matching and minimal back-reflection.

These parameters were measured over a **broad sweep from 75 GHz to 110 GHz**, with special focus on **90–97 GHz**, the operational band of interest for W-band defense applications.

6.4 Key Observations

Observation Point Al6061 Structure

Brass Structure

Surface Roughness (Ra) Higher (Visible Burrs, Unpolished) Lower, smoother finish

Return Loss (S11) Less negative; indicates mismatch Deeper dip; indicates better impedance match

Insertion (S21)	Loss	Higher loss due to roughness and discontinuity	Lower loss; better material conductivity
--------------------	------	--	--

Simulation Agreement	Deviations observed	Close match with CST results
----------------------	---------------------	------------------------------

The experimental S-parameter trends closely followed the **simulation data**, especially in the **Brass prototype**. This correlation confirmed that:

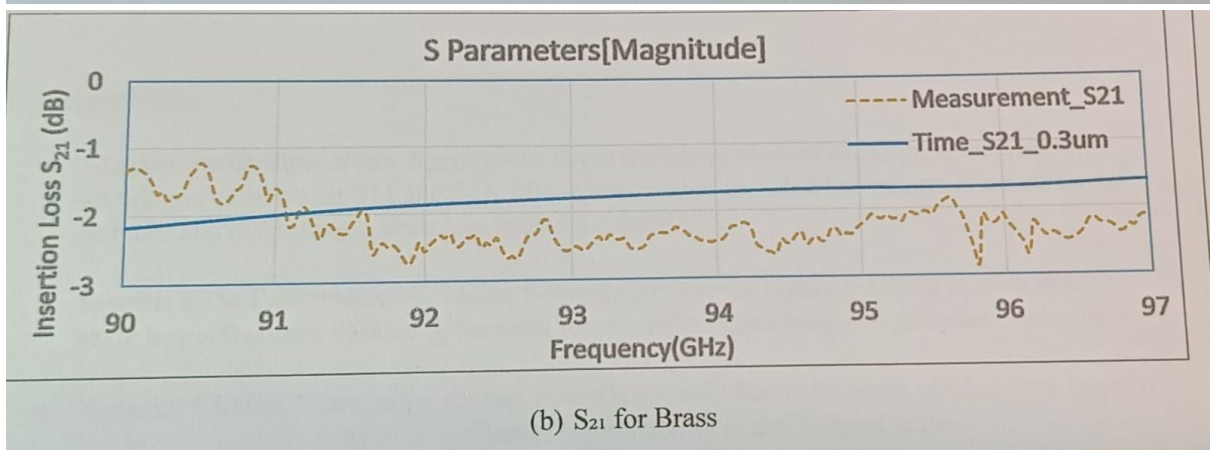
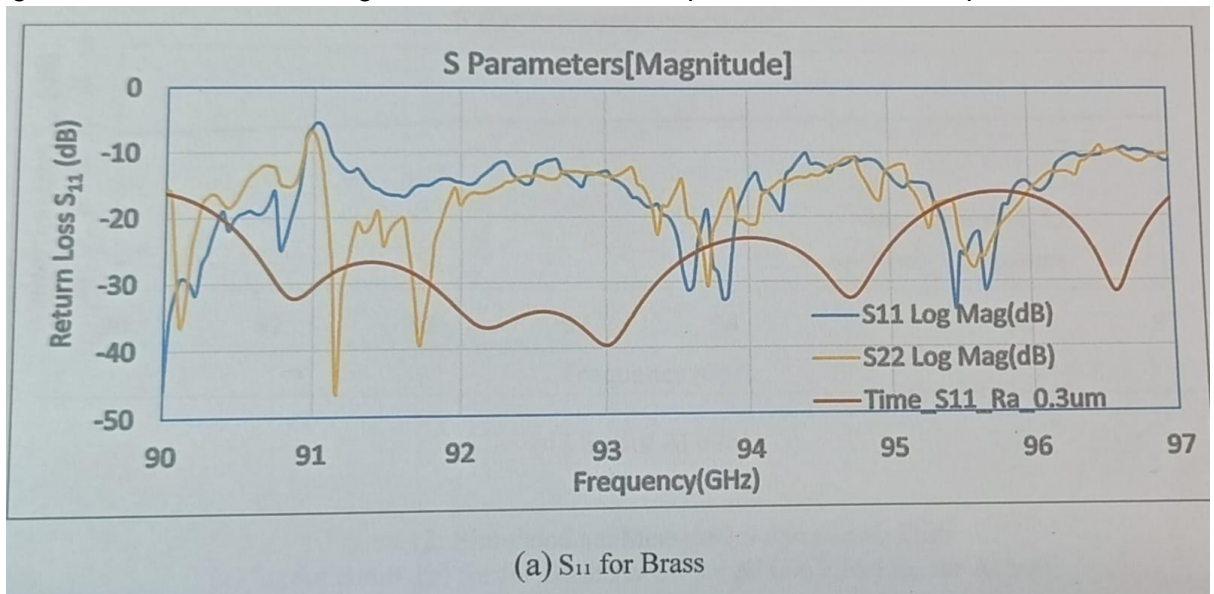
- The **CST model parameters** (including roughness modeling) were accurate
- Surface roughness and flange flatness are **critical loss contributors**
- Cold-test setups are effective in **qualifying structure design before beam-based testing**

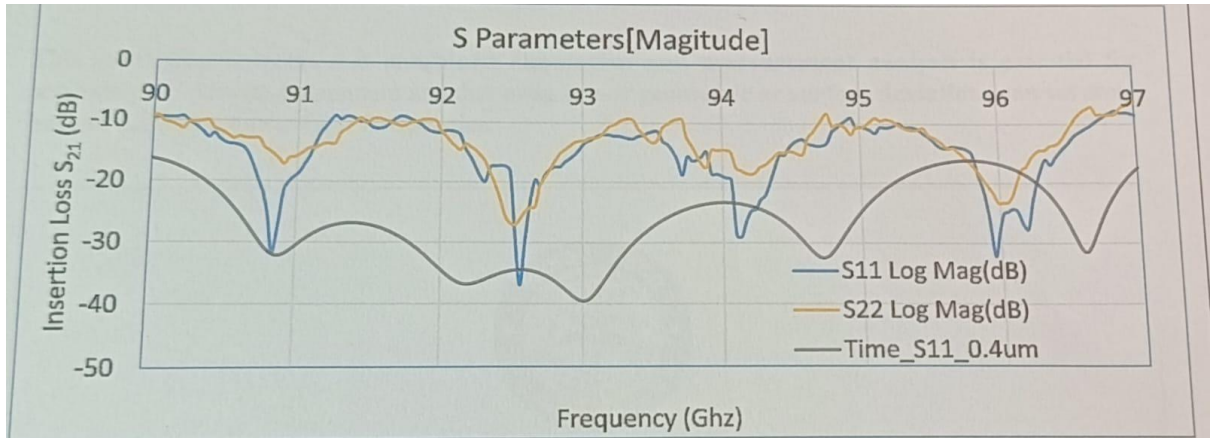


Experimental setup photograph with PNA and WR-10 extenders

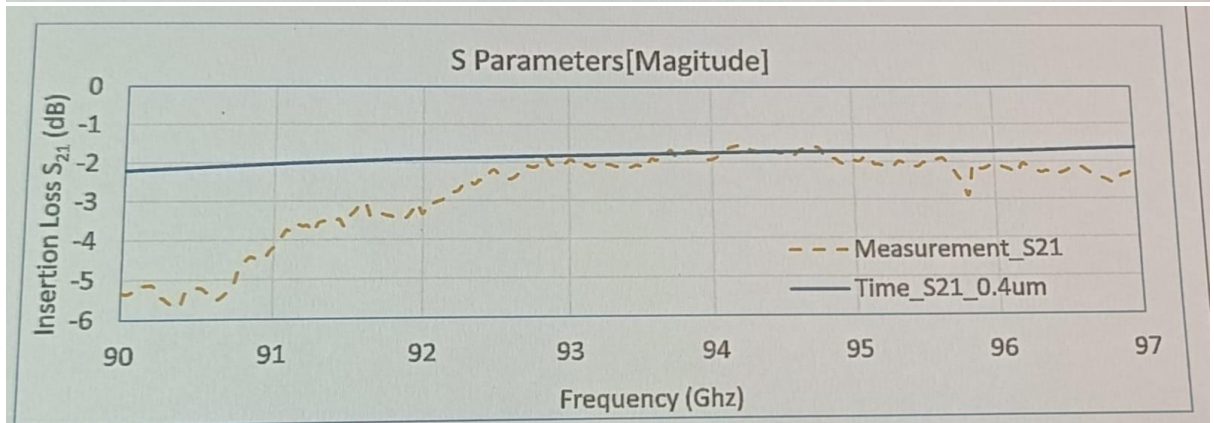
7. RESULTS AND DISCUSSION

- Gained end-to-end exposure from CAD modeling, material machining, parametric simulations, to experimental testing of mm-wave devices.
- Developed proficiency in CST Studio Suite solvers, eigenmode interpretation, and cold test data validation.
- Acquired hands-on understanding of RF losses, fabrication tolerance impacts, and surface quality assessment techniques (Surftest, VMM).
- Identified limitations of conventional CNC machining at W-band and built technical groundwork for transitioning to microfabrication-compatible materials and processes.





(c) S_{11} for Al 6061



(d) S_{21} for Al 6061

Summary

This project integrated mechanical and RF domains to validate the design and manufacturing feasibility of W-band FWTWTs. The groundwork established with Brass and Al6061 structures enabled S-parameter benchmarking and offered insights into machining challenges. The simulation-measurement match provided confidence for further development.

The courses Engineering Electromagnetics (by Dr. Abhishek Jha) and RF Transceiver Design played a critical role. They not only provided theoretical foundations in wave propagation and device modeling but also created networking opportunities that led to the DRDO internship. Interactions with Dr. Jha helped me become proficient in CST tools, understand microwave tube fundamentals, and execute simulations effectively. This support contributed to a strong technical impression at MTRDC, showcasing IIT Tirupati's capabilities.